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03/08/2022

RM-G2
CODE-A

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Corporate Office : Aakash Tower, 8, Pusa Road, New Delhi-110005, Ph.011-47623456

MM : 720

FORTNIGHTLY TEST SERIES

Time : 3 hrs. 20 min

(*for* NEET-2023)

Test – I

Topics covered :

Physics : Physical World, Units and Measurements, Motion in a Straight Line

Chemistry : Some Basic Concepts of Chemistry

Botany : Cell : The Unit of Life

Zoology : Structural Organisation in Animals–Animal Tissues

Instructions :

(i) There are two sections in each subject, i.e. Section-A & Section-B. You have to attempt all 35 questions from Section-A & only 10 questions from Section-B out of 15.

(ii) Each question carries 4 marks. For every wrong response 1 mark shall be deducted from the total score. Unanswered / unattempted questions will be given no marks.

(iii) Use blue/black ballpoint pen only to darken the appropriate circle.

(iv) Mark should be dark and completely fill the circle.

(v) Dark only one circle for each entry.

(vi) Dark the circle in the space provided only.

(vii) Rough work must not be done on the Answer sheet and do not use white-fluid or any other rubbing material on the Answer sheet.

PHYSICS

Choose the correct answer:

SECTION-A

1. The range of gravitational forces is
   (1) 10^{-15} m (2) 10^{-16} m
   (3) 10^{15} m (4) Infinite
2. Choose the incorrect matching.
   (1) 1 nm = 10^{-9} m
   (2) 1 pm = 10^{-12} m
   (3) 1 mg = 10^{-3} kg
   (4) Both (1) and (2)
3. The scientific principle involved in TV broadcast is
   (1) Amplification by population inversion
   (2) Superconductivity
   (3) Fluid dynamics
   (4) Principle of communication system using EM wave
4. Classical physics deals with
   (1) Microscopic phenomenon
   (2) Macroscopic phenomenon
   (3) Mesoscopic phenomenon
   (4) Both (1) and (3)
5. Choose the correct pair of name of scientist and their discovery among the following.
   (1) H. Hertz – Telephone
   (2) Graham Bell – Electric transformer
   (3) C.V. Raman – Scattering of light
   (4) Kepler – Barometer
6. 1° is equal to
   (1) $\frac{\pi}{180}$ rad (2) 60′
   (3) $\frac{180}{\pi}$ rad (4) Both (1) and (2)
7. Which of the following is correct representation of SI unit of force?
   (1) J (2) newton-meter
   (3) N (4) meter/second^2
8. The radius of a circle is 0.020 m. The number of significant figures is
   (1) 1 (2) 2
   (3) 3 (4) 4
9. The quantity having both unit and dimensions, is
   (1) Velocity (2) Plane angle
   (3) Strain (4) Specific gravity
10. The value of $(10004)^{1/4}$ according to binomial approximation is
    (1) 10.001 (2) 10.04
    (3) 10.004 (4) 10.01
11. Which of the following measurements, is most precise?
    (1) 0.002 m (2) 0.02 m
    (3) 0.2 m (4) 0.0002 m
12. The length of a rod is measured as 110.32 cm. The value upto 3 significant figures is
    (1) 111 cm (2) 1.10×10^2 cm
    (3) 110.3 cm (4) 1.11×10^2 cm
13. Dimensional formula of power is given by
    (1) $[ML^2T^{-2}]$ (2) $[MLT^{-2}]$
    (3) $[ML^2T^{-3}]$ (4) $[ML^{-2}T^{-3}]$
14. If $A = (10.0 \pm 0.1)$ and $B = (5.0 \pm 0.1)$, then $2A - B$ is equal to
    (1) (15.0 ± 0.3) (2) (15.0 ± 0.1)
    (3) (10.0 ± 0.3) (4) (20.0 ± 0.3)
15. Choose the correct pair of physical quantity and their SI unit.
    (1) Electric current – A
    (2) Solid angle – rad
    (3) Strain – Pa
    (4) Relative density – kg/m^3
16. A physical quantity p is given as $p = \frac{A^3}{\sqrt{BC}}$ and the percentage errors in the measurements of A, B and C are 1%, 2% and 4% respectively. The maximum percentage error in the calculation of p is
    (1) 4% (2) 6%
    (3) 8% (4) 10%

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17. The number of significant zeros present in the measured value 0.0040200 is

(1) 8 (2) 3

(3) 7 (4) 5

18. In the addition of three masses 1.4 g, 4.24 g and 7.432 g has x significant figures. The value of x is

(1) 2 (2) 3

(3) 4 (4) 5

19. In the morning, temperature of a room is measured as (31 ± 1)°C and in noon it is measured as (44 ± 2)°C. The rise in temperature will be

(1) (13 ± 1)°C (2) (13 ± 2)°C

(3) (13 ± 3)°C (4) (13 ± 4)°C

20. Zero error in an instrument is a

(1) Random error

(2) Systematic error

(3) Least count error

(4) Personal error

21. The pair of quantities having same dimensions is

(1) Displacement, Focal length

(2) Tension, Surface tension

(3) Power, Time

(4) Force, Impulse

22. Which of the following is a dimensionless constant?

(1) Gravitational constant

(2) Boltzmann's constant

(3) Magnification produced by mirror

(4) Stefan's constant

23. A physical quantity z is given by the relation $z = A\sin(kx - \omega t)$. If x and t denotes displacement and time respectively, then dimension of $\left[\frac{\omega}{k}\right]$ is

(1) [LT] (2) $[LT^{-2}]$

(3) $[L^{-1}T]$ (4) $[LT^{-1}]$

24. 100 J of energy is equivalent to K ergs (in CGS system), The value of K is

(1) 10^7 (2) 10^8

(3) 10^9 (4) 10^5

25. If force (F), velocity (u) and time (T) are taken as fundamental quantities, then dimension of length is

(1) [F u T] (2) $[F^0 u T]$

(3) $[F u^{-1} T^{-1}]$ (4) $[F^{-1} u T^{-1}]$

26. Slope of position-time (x-t) graph represents

(1) Average acceleration

(2) Instantaneous velocity

(3) Areal velocity

(4) Instantaneous acceleration

27. A particle moves 10 m in the east direction and then 5 m to west direction. The magnitude of displacement covered by the particle is

(1) 5 m (2) 10 m

(3) 15 m (4) 7.5 m

28. A car is moving with speed 100 km h^{-1}. The speed (in m s^{-1}) is equal to

(1) 100 (2) 27.78

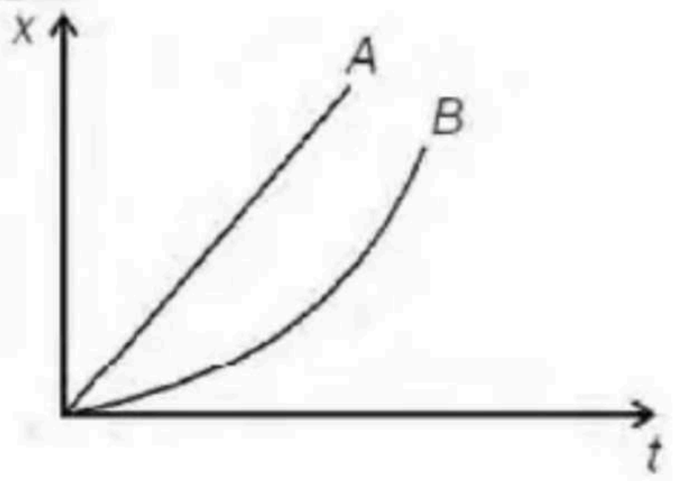
(3) 25.4 (4) 50.78

29. The differentiation of a function $y = x^3$ w.r.t. x is

(1) $2x^3$ (2) $3x^2$

(3) $3x^3$ (4) $6x$

30. The displacement-time (x-t) graph of two moving objects A and B are shown in the figure. Choose the incorrect statement.



(1) A is moving with constant velocity

(2) B is moving with increasing speed

(3) The acceleration of B may be constant

(4) A is moving with non-zero constant acceleration

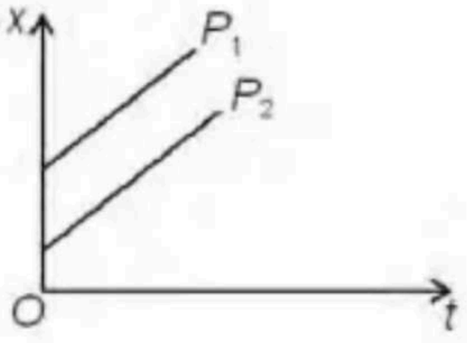
31. Two cars are moving in a straight line in same direction with same speed of 70 km h^{-1}. The relative velocity of one car with respect to other is

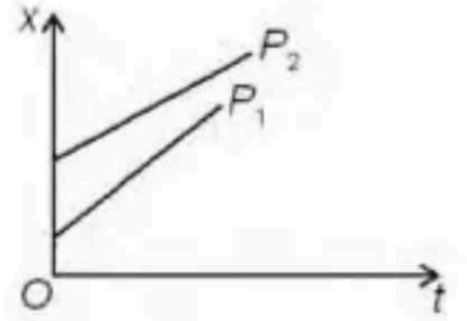
(1) 70 km h^{-1} (2) 140 km h^{-1}

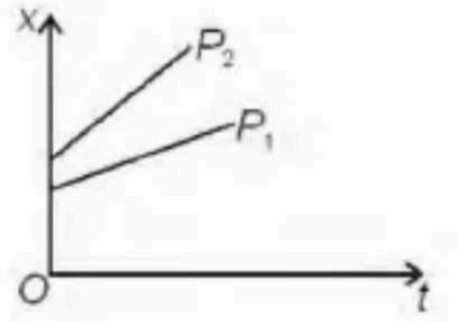
(3) 35 km h^{-1} (4) Zero

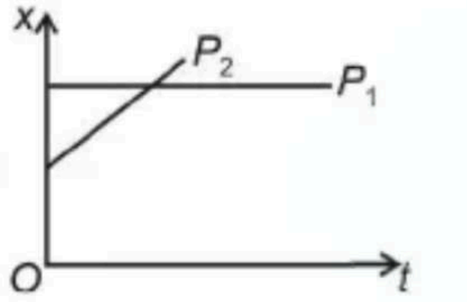
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32. Two persons P_1 and P_2 are walking with speeds v_1 and v_2 respectively along east (+ve x) direction. If $v_1 > v_2$ then, which of the following position-time graph is correctly showing velocities of P_1 and P_2?

(1) 

(2) 

(3) 

(4) 

33. A body is moving with variable acceleration (α) along a straight line. The average acceleration of the body in time interval $t = 0$ to $t = t_0$ is

(1) $\frac{\alpha t_0}{2}$ (2) $\frac{\alpha t_0}{4}$

(3) $\frac{\int_0^{t_0} \alpha dt}{t_0}$ (4) Zero

34. A person travels along a straight road for first $\frac{t}{5}$ time with speed v_1 and for next $\frac{4t}{5}$ time with a speed v_2. The mean speed v is given by

(1) $v = \frac{v_1 + v_2}{2}$ (2) $v = \frac{v_1 + 4v_2}{5}$

(3) $\frac{1}{v} = \frac{1}{5v_1} + \frac{4}{5v_2}$ (4) $v = \frac{2}{5}\sqrt{v_1 v_2}$

35. A body in one dimensional motion has zero speed at an instant. At that instant, body must have

(1) Zero velocity

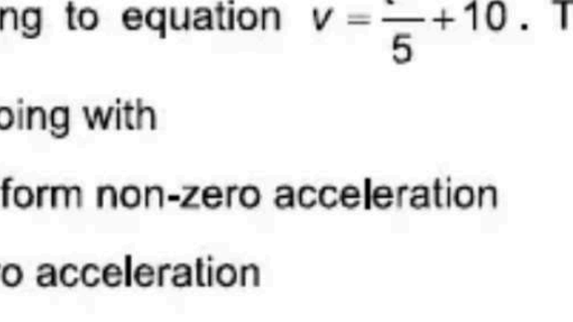
(2) Non-zero velocity

(3) Zero acceleration

(4) Non-zero acceleration

SECTION-B

36. The position-time graph for a particle moving along a straight line is shown in the figure. The total distance travelled by it in time $t = 0$ to $t = 6$ s is



(1) Zero (2) 20 m

(3) 40 m (4) 80 m

37. A ball is dropped from the top of a building of height 100 m at time $t = 0$. At same instant, another ball is thrown upwards with speed 50 m/s from the bottom of the building. The time at which balls will meet is

(1) 2 s (2) 4 s

(3) 3 s (4) 2.5 s

38. The velocity of a body depends on the time according to equation $v = \frac{t^2}{5} + 10$. The body is undergoing with

(1) Uniform non-zero acceleration

(2) Zero acceleration

(3) Uniform retardation

(4) Non-uniform acceleration

39. An object thrown vertically up from the ground passes the height 10 m twice in the time interval of 10 s. The time of flight is (g = 10 m/s^2)

(1) $6\sqrt{3}$ s (2) $3\sqrt{3}$ s

(3) $2\sqrt{3}$ s (4) $\sqrt{3}$ s

40. The distance travelled by a particle starting from rest and moving with an acceleration 2 m/s^2 in 5th second is

(1) 4 m (2) 9 m

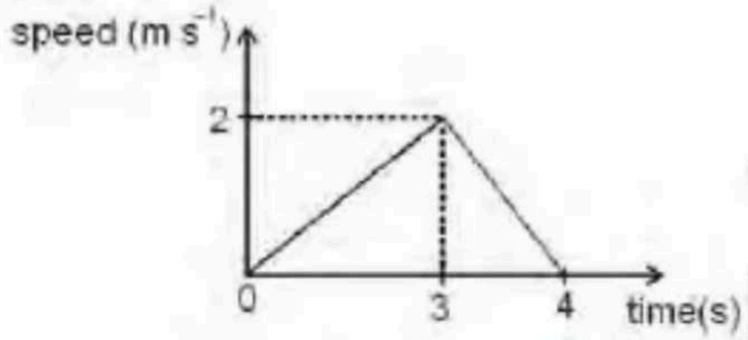
(3) 5 m (4) 25 m

41. The motion of a particle along a straight line is described by equation $x = 2 + 12t - t^2$, where x is in m and t is in s. The retardation of the particle when its velocity becomes zero is

(1) 2 m/s^2 (2) 12 m/s^2

(3) 4 m/s^2 (4) 8 m/s^2

42. The speed-time graph of a particle moving along a straight line is shown in the figure. The distance covered by the particle from time $t = 0$ to $t = 4$ s is



(1) 2 m (2) 4 m

(3) 6 m (4) 3 m

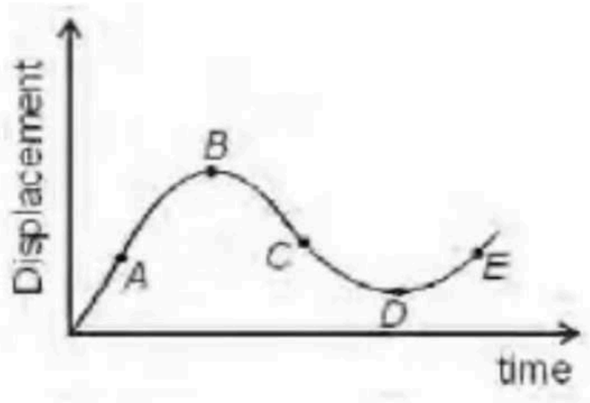
43. Motion of a particle is given by the equation

$s = (2t^3 + 3t^2 + 4t + 2)$ m, where s is displacement in meter and t is time in second. The value of acceleration of particle at time $t = 1$ s is

(1) 12 m/s^2 (2) 18 m/s^2

(3) 6 m/s^2 (4) Zero

44. The displacement-time graph of a moving particle is shown in the figure. The instantaneous velocity of the particle is negative at point



(1) A

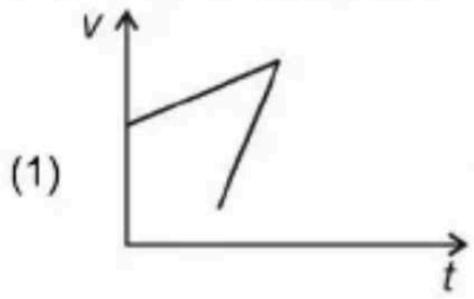
(2) C

(3) B

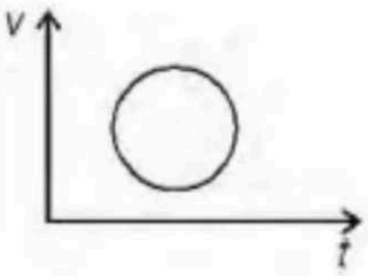
(4) D

45. Which of the following speed-time (v-t) graph is physically not possible?

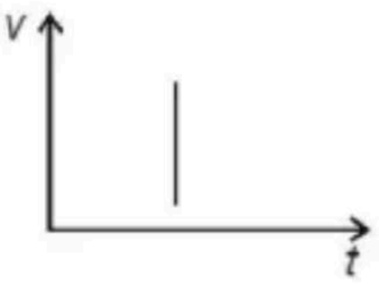
(1)



(2)



(3)



(4) All of these

46. The dimensions of $\frac{\alpha}{\beta^2}$ in $P\alpha = F\beta e^{-\beta t}$ (Where P is pressure, F is force and t is time) is

(1) [L^{-2}T^{-1}] (2) [L^{2}T^{-1}]

(3) [L^{2}T] (4) [L^{2}T^{-2}]

47. While measuring density of a body its mass was found as (20.0 ± 0.4) kg contained in a volume of (5.00 ± 0.20) m^3.

The density of this body under permissible limit of error is

(1) (5 ± 0.65) kg m^{-3} (2) (4 ± 0.15) kg m^{-3}

(3) (4 ± 0.24) kg m^{-3} (4) (4 ± 0.65) kg m^{-3}

48. Distance travelled by particle in 5 seconds for a motion whose velocity varies as $V = (2t - 4)$ is

[v in m/s and t in s]

(1) 5 m (2) 13 m

(3) 9 m (4) 21 m

49. Two particles A and B (such that mass of A is greater than B) are thrown vertically upwards and downwards respectively with same speed from a tower. Choose the correct statement.

(1) Distance covered by both of them is same

(2) Velocity of both particles is same while hitting the ground

(3) Kinetic energy of both particles is same while hitting the ground

(4) Speed of A is smaller than speed of B while hitting the ground

50. Two trains A and B of equal lengths running at 72 km/h and 90 km/h respectively. If A accelerates at 2 ms^{-2}, then it overtakes B in 20 seconds.

The length of both trains is

(1) 150 m (2) 100 m

(3) 300 m (4) 50 m

CHEMISTRY

SECTION-A

51. For reaction, A + 2B → C, 5 moles of A and 12 moles of B will produce maximum _____ of C.
(1) 5 mole (2) 6 mole
(3) 10 mole (4) 12 mole

52. Number of oxygen atoms present in 22 g of CO_2 is
(1) 2 N_A atoms (2) 0.5 N_A atoms
(3) N_A atoms (4) 1.5 N_A atoms

53. The molarity of 24.5% (w/w) H_2SO_4 solution having density 1.2 g/ml is
(1) 1 M (2) 3 M
(3) 1.2 M (4) 2.4 M

54. 6.022×10^{23} molecules of urea are present in 1000 g water. The molality of urea solution is
(1) 0.1 m (2) 0.01 m
(3) 10 m (4) 1 m

55. If mass percentage of zinc in an enzyme is 0.1308% then the minimum molecular mass of the enzyme will be (Atomic mass of zinc = 65.4 u)
(1) 40000 u (2) 50000 u
(3) 20000 u (4) 10000 u

56. The mass percentage of oxygen in $CaCO_3$ is
(1) 40% (2) 12%
(3) 16% (4) 48%

57. The molarity of 2 N H_2SO_4 solution is
(1) 1 M (2) 2 M
(3) 0.5 M (4) 1.5 M

58. For complete oxidation of 2 litre of NO(g) to NO_2(g) at STP, the volume of O_2(g) required at STP is
(1) 0.5 L (2) 2 L
(3) 1 L (4) 1.5 L

59. Number of P atoms present in 0.5 g molecules of P_4 is
(1) N_A (2) 2 N_A
(3) 0.5 N_A (4) 1.5 N_A

60. The maximum number of molecules are present in
(1) 4 g H_2 (2) 28 g N_2
(3) 0.5 mole O_2 (4) 11 g CO_2

61. 11.1 g of $CaCl_2$ is present in 200 mL of an aqueous solution. The molarity of chloride ion in the solution is (Atomic mass of Ca = 40 u and Cl = 35.5 u)
(1) 1.5 M (2) 0.5 M
(3) 2 M (4) 1 M

62. How many significant figures are in 0.03412?
(1) 1 (2) 5
(3) 6 (4) 4

63. The incorrect statement according to Dalton's atomic theory is
(1) Compounds are formed when atoms of different elements combine in a fixed ratio
(2) Atoms of different elements differ in mass
(3) Atoms of a given element are identical in all respects
(4) Atoms can be created in a chemical reaction

64. Which of the following combinations does not illustrate the law of multiple proportions?
(1) H_2O, H_2O_2 (2) SO_2, SO_3
(3) CS_2, CO_2 (4) N_2O_4, NO_2

65. The mass of one molecule of H_2O is
(1) 18 N_A g (2) $\frac{18}{N_A}$ g
(3) 18 g (4) $\frac{N_A}{18}$ g

66. The empirical formula of a hydrocarbon containing 80% carbon is
(1) CH_4 (2) CH_3
(3) C_2H_3 (4) CH_2

67. 2.24 L of methane is present in a closed container at STP. The number of atoms of methane in the container is
(1) 6.02×10^{23} (2) 3.01×10^{23}
(3) 3.01×10^{22} (4) 6.02×10^{21}

68. Which of the following is not an SI unit?
(1) gram
(2) mole
(3) second
(4) metre

69. For which reaction, Gay Lussac's law is not applicable?

(1) $2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$

(2) $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$

(3) $2NH_3(g) \rightarrow N_2(g) + 3H_2(g)$

(4) $2H_2(g) + O_2(g) \rightarrow 2H_2O(g)$

70. Which among the following is independent of temperature?

(1) Molality (2) Molarity

(3) Normality (4) (w/v) %

71. The mole fraction of solute in aqueous solution of NaOH having molality one is

(1) 0.042 (2) 0.095

(3) 0.018 (4) 0.032

72. An element X has the following isotopic composition. X^{200} : 90% and X^{202} : 10%. The average atomic mass of the naturally occurring element X is

(1) 201.5 amu (2) 200.8 amu

(3) 200.5 amu (4) 200.2 amu

73. The volume occupied by one mole of water is (density = 1 g/ml)

(1) $\frac{1}{N_A}$ml (2) 18 ml

(3) $\frac{2}{N_A}$ml (4) 9 ml

74. The volume of 0.2 M NaOH required to prepare 0.1 M, 1000 ml NaOH solution is

(1) 250 ml (2) 500 ml

(3) 1000 ml (4) 750 ml

75. The number of electrons present in 1.2 g of O^{2-} is

(1) N_A (2) 0.75 N_A

(3) 10 N_A (4) 0.1 N_A

76. The volume of $O_2(g)$ required to completely burn 16 g CH_4 at STP is

(1) 22.4 L (2) 44.8 L

(3) 11.2 L (4) 5.6 L

77. The molarity of pure water is (density = 1 g/cm^3)

(1) 55.55 M (2) 45.45 M

(3) 30.55 M (4) 18 M

78. A solution is prepared by adding 90 g glucose in 180 g water. The mole fraction of glucose in the solution is

(1) $\frac{1}{20}$ (2) $\frac{1}{16}$

(3) $\frac{1}{15}$ (4) $\frac{1}{21}$

79. Number of Be atoms in 36 u of Be^9 is

(1) 9 (2) 12

(3) 36 (4) 4

80. In different samples of CO_2, mass ratio of carbon to oxygen is found to be 3 : 8. This observation obeys

(1) Law of conservation of mass

(2) Law of multiple proportion

(3) Gay Lussac's law of gaseous volumes

(4) Law of definite proportion

81. In which of the following formula mass is preferred over molecular mass?

(1) KCl (2) C_2H_6

(3) N_2 (4) H_2O

82. Select the incorrect statement among the following.

(1) If mass percentage of various elements present in a compound is known then the empirical formula of compound can be determined

(2) The reactant which gets completely consumed first in the reaction is called limiting reagent

(3) One amu is $\frac{1}{12}$th of the mass of one C-12 atom

(4) Molecular formula represents the simplest whole number ratio of various atoms present in a compound

83. If 30 g of NO_x (g) occupies 22.4 L at STP then the value of x is

(1) 2 (2) 1

(3) 3 (4) $\frac{1}{2}$

84. Mole(s) of O_2 required to produce 160 g of SO_3 from SO_2 is

(1) 0.5 (2) 1

(3) 2.5 (4) 2

85. Equal masses of $O_2(g)$ and $H_2(g)$ are taken in a closed container at STP. The ratio of the moles of gases O_2 : H_2 will be

(1) 1 : 8 (2) 1 : 4

(3) 1 : 16 (4) 1 : 10

SECTION-B

86. Incorrect statement among the following is

(1) Isotopes of an element have equal relative abundances

(2) Many naturally occurring elements exist as more than one isotopes

(3) Atomic mass of an element is average atomic mass of its different isotopes

(4) The percentage composition of elements in a compound is independent of its source

87. Mass of one amu in kg is

(1) 1.66×10^{-27} (2) 6.022×10^{-27}

(3) 6.022×10^{-23} (4) 1.66×10^{-24}

88. The ratio of molecular formula mass to empirical formula mass of acetic acid is

(1) 1 : 2 (2) 2 : 1

(3) 2 : 3 (4) 3 : 2

89. Mass of 25% (w/w) HCl solution required to decompose 50 g $CaCO_3$ is

(1) 36.5×2 g (2) 36.5 g

(3) 36.5×0.5 g (4) 36.5×4 g

90. Moles of oxygen atoms present in 0.1 mole of $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$ is

(1) 0.8 (2) 0.9

(3) 1.4 (4) 1.7

91. Which of the following molecules contains maximum percentage of carbon by mass?

(1) C_6H_6 (2) C_2H_6

(3) C_2H_4 (4) C_3H_4

92. How many grams of concentrated nitric acid solution should be used to prepare 200 mL of 3.0 M HNO_3 solution?

(Concentrated nitric acid is 70% (w/w) HNO_3)

(1) 36 g

(2) 72 g

(3) 60 g

(4) 54 g

93. 3.01×10^{23} molecules of glucose are present in 500 ml of solution. The molarity of the solution is

(1) 0.5 M (2) 1 M

(3) 2 M (4) 1.5 M

94. What is the percentage purity of impure magnesium carbonate sample whose 8.4 g sample on decomposition gives 2.2 g CO_2?

(1) 25% (2) 100%

(3) 50% (4) 75%

95. When 100 ml of 17% (w/v) solution of $AgNO_3$ is mixed with 100 ml of 5.85% (w/v) NaCl solution, the mass of AgCl formed will be (Ag = 108, N = 14, Na = 23, Cl = 35.5)

(1) 14.35 g (2) 27.5 g

(3) 8.35 g (4) 11.35 g

96. If temperature of water bath is 150°F then its value in °C will be

(1) 25.25°C (2) 65.56°C

(3) 72.1°C (4) 55°C

97. Mass of $CaCO_3$ required to react completely with 200 mL of 4 M HCl solution is

(1) 20 g (2) 25 g

(3) 50 g (4) 40 g

98. A container contains 20 L of water. The volume of water in m^3 will be

(1) 20×10^{-4} m^3 (2) 2×10^{-2} m^3

(3) 2×10^{-3} m^3 (4) 20 m^3

99. Prefix used for 10^{-12} in SI system is

(1) Nano (2) Femto

(3) Micro (4) Pico

100. **Statement I:** Zeros between two non-zero digits are significant.

Statement II: Zeros preceding to first non-zero digit are not significant.

In light of the above statements, choose the correct answer.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

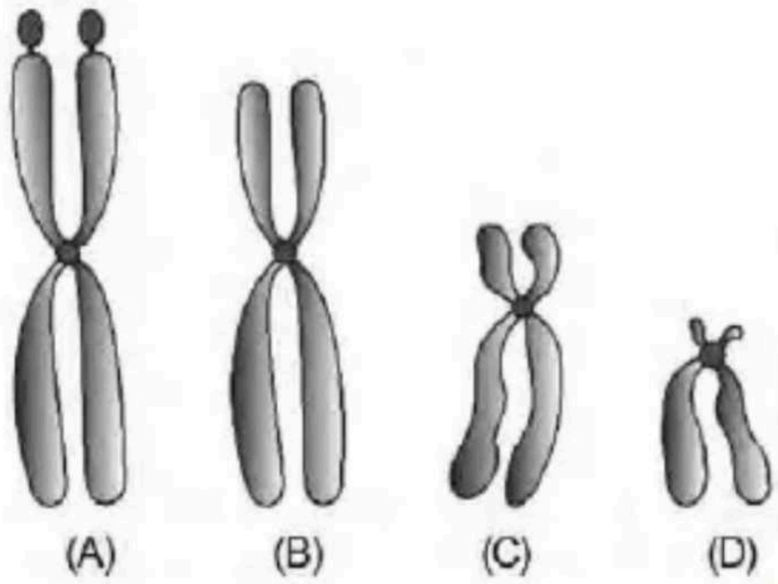
(4) Statement I is incorrect but statement II is correct

BOTANY

SECTION-A

101. Many membrane bounded minute vesicles called microbodies that contain various enzymes are present in both
(1) Bacteria and plant cells
(2) Animal and plant cells
(3) Bacteria and animal cells
(4) PPLO and cyanobacteria

102. Which of the following chromosomes are SAT-chromosome (having satellite) and acrocentric chromosome, respectively?



(1) (A) and (B) (2) (B) and (C)
(3) (A) and (D) (4) (A) and (C)

103. Chromatin material is made up of
(1) RNA and basic histones protein only
(2) DNA, RNA, histone and non-histone proteins
(3) DNA, RNA and histones only
(4) DNA and histones only

104. Which of the following is the site of ribosomal RNA synthesis?
(1) Nucleolus (2) Lysosome
(3) Cytoplasm (4) Golgi apparatus

105. How many centrioles are present in a centrosome?
(1) One (2) Two
(3) Three (4) Four

106. The centriole-like structure from which cilia and flagella emerge in eukaryotes is called
(1) Basal body (2) Hook
(3) Axoneme (4) Spindle fibres

107. The structure which work like oars causing movement of either cell or surrounding fluid is
(1) Flagella
(2) Cilia
(3) Central hub
(4) Massule

108. Choose the **incorrect** statement w.r.t. to eukaryotic flagella.
(1) Hair-like outgrowths of cell membrane
(2) Elongated structure and responsible for cell movement
(3) It is structurally same as prokaryotic flagella
(4) Possess a number of microtubules running parallel to the long axis

109. A living cell was first seen by
(1) Robert Brown
(2) Anton von Leeuwenhoek
(3) Schleiden
(4) Schwann

110. Which of the following non-membrane bound structure is found in eukaryotic cells?
(1) Vacuole (2) Lysosome
(3) Ribosome (4) ER

111. Match the following and choose the **correct** option.

a. Smallest cell	(i) Ostrich's egg
b. Longest isolated cell	(ii) *Mycoplasma*
c. Largest cell	(iii) Nerve cell

(1) a(ii), b(iii), c(i)
(2) a(iii), b(ii), c(i)
(3) a(ii), b(i), c(iii)
(4) a(i), b(iii), c(ii)

112. Constituents of cytoskeleton are composed of
(1) Cellulose
(2) Proteinaceous filaments
(3) Chitin
(4) Cellulose and hemicellulose

113. Read the following statements and select which are **not** the postulates of cell theory.

(a) Cell differ greatly in size, shape and activities.

(b) All living organisms are composed of cells and products of cells.

(c) The cytoplasm is the main arena of cellular activities.

(d) All cells arise from pre-existing cells.

(1) (a), (d) (2) (b), (d)

(3) (a), (c) (4) (c), (d)

114. The main chemical constituent of fungal cell wall is

(1) Pectin (2) Hemicellulose

(3) Chitin (4) Peptidoglycan

115. The genetic material is naked and **not** surrounded by a nuclear membrane in

(1) PPLO

(2) WBC

(3) Columnar epithelial cell

(4) Mesophyll cell

116. Which of the following statements is **correct** about Gram positive and Gram negative bacteria?

(1) Gram stain is not taken by Gram positive but taken by Gram negative

(2) Gram stain is taken by both Gram positive and Gram negative

(3) Gram stain is taken by Gram positive but not taken by Gram negative

(4) Gram stain is not taken by both Gram positive and Gram negative

117. When glycocalyx is thick and tough, it is called

(1) Capsule (2) Slime layer

(3) Cell wall (4) Cell membrane

118. Function which is **not** associated with mesosome is

(1) DNA replication (2) Secretion

(3) Respiration (4) Photosynthesis

119. Pigment containing membranous extensions are present into the cytoplasm of

(1) Virus (2) Cyanobacteria

(3) Fungi (4) Animal cell

120. The function of fimbriae in bacteria is

(1) To help in attaching to host tissue or to rocks

(2) Motility

(3) DNA replication

(4) Secretion process

121. A polysome is

(1) Several m-RNAs attached to a single ribosome

(2) Several ribosomes attached to a single m-RNA

(3) Association of single m-RNA and a single ribosome

(4) Association of several ribosomes together

122. Reserve materials in prokaryotic cells are stored in cytoplasm in the form of inclusion bodies which are

(1) Membrane bound structures

(2) Non-membrane bound structures

(3) Always associated with plasma membrane of cell

(4) Rich in starch granules only

123. Choose the **incorrect** statement regarding eukaryotic cells.

(a) Presence of extensive compartmentalisation of cytoplasm

(b) Presence of an organised nucleus having a nuclear envelope

(c) Cytoskeletal structures are absent

(d) Genetic material is organised into circular DNA

(1) (a) and (b) only

(2) (b) and (c) only

(3) (c) and (d) only

(4) All (a), (b), (c) and (d)

124. Phosphoglycerides are the main constituent of

(1) Lipid component of cell membrane

(2) Protein component of cell membrane

(3) Lipid component of cell wall

(4) Protein component of cell wall

125. The membrane proteins are classified as integral or peripheral. The basis of this classification is
(1) Quasi-fluid nature
(2) Biochemical investigation
(3) Ease of extraction
(4) Requirement of energy for transport of material

126. The fluidity of cell membrane is measured by
(1) The ability of lateral movement of proteins
(2) Cell growth
(3) The formation of intercellular junctions
(4) Karyokinesis

127. Match the following columns and choose the **correct** answer.

	Column I		Column II
a.	Passive transport	(i)	Movement of ions against concentration gradient
b.	Active transport	(ii)	Movement of water by diffusion through semi-permeable membrane
c.	Osmosis	(iii)	Simple diffusion along concentration gradient

(1) a(iii), b(ii), c(i) (2) a(iii), b(i), c(ii)
(3) a(ii), b(iii), c(i) (4) a(i), b(iii), c(ii)

128. Which of the following is **not** the functions of cell wall?
(1) Gives shape to the cells
(2) Protects from mechanical damage and infection
(3) Regulates the transport of the molecules across it
(4) Helps in cell to cell interaction

129. Middle lamella, a layer which holds the neighbouring cells together is mainly made up of
(1) Calcium carbonate (2) Calcium pectate
(3) Galactans (4) Mannans

130. The functions are co-ordinated in organelles
(1) Peroxisome and ribosome
(2) Mitochondria and ER
(3) Chloroplast and ER
(4) ER and Golgi body

131. Rough endoplasmic reticulum is frequently observed in the cells actively involved in
(1) Protein synthesis and secretion
(2) Protein breakdown and absorption
(3) Oxidation of carbohydrates
(4) DNA duplication

132. Synthesis of steroidal hormones occurs in
(1) SER (2) RER
(3) Golgi apparatus (4) Mitochondria

133. Chromosome in which centromere is present at the terminus
(1) Is called metacentric chromosome
(2) Is J-shaped
(3) Appears to have only one arm
(4) Does not contain DNA at all

134. The cytoskeletal element which helps in pseudopodia formation is
(1) Microtubules
(2) Centriole
(3) Intermediate filament
(4) Microfilament

135. Which of the following cell organelles is related to the packaging of material?
(1) Nucleus (2) Golgi apparatus
(3) Mitochondria (4) Chloroplast

SECTION-B

136. The enzymes of lysosomes are optimally active at
(1) Neutral pH (2) Acidic pH
(3) Moderate basic pH (4) Highly basic pH

137. Tonoplast is the
(1) Single membrane around lysosome
(2) Double membrane complex around lysosome
(3) Single membrane around sap-vacuole
(4) Double membrane complex around sap-vacuole

138. Choose the **incorrect** statement about mitochondria.
(1) Site of anaerobic respiration
(2) Produce cellular energy in the form of ATP
(3) Are called 'power house' of cell
(4) Possess single circular DNA molecule in their matrix

139. Match the plastids with their concerning colour and choose the **correct** option.

a. Leucoplasts	(i) Yellow, orange and red coloured
b. Chromoplast	(ii) Green coloured
c. Chloroplasts	(iii) Colourless

(1) a(ii), b(i), c(iii)
(2) a(iii), b(i), c(ii)
(3) a(iii), b(ii), c(i)
(4) a(ii), b(iii), c(i)

140. The pigments carotene and xanthophylls are soluble in
(1) Fat (2) Water
(3) Protein (4) Carbohydrate

141. Read the following statements and choose the **correct** option.
(a) Amyloplasts store starch
(b) Aleuroplasts store proteins
(c) Elaioplasts store both starch and proteins
(d) Amyloplasts, elaioplasts and aleuroplasts all are types of plastids having photosynthetic pigments
(1) Only (a) and (b) are correct
(2) Only (c) is incorrect
(3) Only (d) is incorrect
(4) Only (c) and (d) are correct

142. The space limited by the inner membrane of chloroplast is called
(1) Intermembrane space
(2) Stroma
(3) Lumen
(4) Thylakoid

143. How many microtubules are found in the transverse section of cilia and flagella?
(1) 1 (2) 9
(3) 20 (4) 27

144. Contractile vacuole is important for excretion in
(1) *Chlamydomonas*
(2) *Amoeba*
(3) Fungi
(4) All plant cells

145. Plasmid is related to all, **except**
(1) Extra chromosomal DNA
(2) Smaller than genomic DNA
(3) Linear DNA
(4) Confers certain unique phenotypic character to bacteria

146. In prokaryotes, a specialised differentiated form of cell membrane is called
(1) Tubules
(2) Mesosomes
(3) Capsule
(4) Filament

147. Which among the following structure helps in the motility of prokaryotic cell?
(1) Pili (2) Flagella
(3) Fimbriae (4) Cilia

148. Which organelle is found within the organelle of cell?
(1) Ribosome
(2) Lysosomes
(3) Golgi body
(4) Mitochondria

149. Higher plant cells and animal cells are different as the former possess all, **except**
(1) Cell wall
(2) Large vacuole
(3) Plastids
(4) Centrioles

150. A few chromosomes have non-staining A at a constant location. This gives the appearance of small fragment called the satellite.
Select the **correct** option for A.
(1) Centromere
(2) Primary constriction
(3) Secondary constriction
(4) Kinetochore

ZOOLOGY

SECTION-A

151. A group of similar cells, having same origin and performing a specific function form a/an
(1) Organ (2) Organ system
(3) Body (4) Tissue

152. Read the following statements and choose the **correct** option.

Statement A : The structure of a cell varies according to its function.

Statement B : Depending on structure and function, the animal tissues have been broadly classified into four general types.

(1) Both statements are incorrect
(2) Only statement A is correct
(3) Only statement B is correct
(4) Both statements are correct

153. Match the following given structures with their characteristic features and choose the **correct** option.

a. Tight junctions	(i) Perform cementing to keep neighbouring cells together
b. Adhering junctions	(ii) Transmit impulses through chemicals to adjacent cell
c. Gap junctions	(iii) Establish a barrier to prevent leakage of substances across the epithelial cells
d. Synaptic junctions	(iv) Channels to facilitate communication between adjacent cells

(1) a(iv), b(iii), c(i), d(ii) (2) a(ii), b(iv), c(i), d(iii)
(3) a(iv), b(ii), c(i), d(iii) (4) a(iii), b(i), c(iv), d(ii)

154. In humans, ciliated epithelial cells are required to move particles or mucus in a specific direction. These cells are mainly present in
(1) Bronchioles (2) Eustachian tube
(3) Pancreatic duct (4) Salivary duct

155. The epithelium which is also known as pavement epithelium is
(1) Simple cuboidal (2) Simple squamous
(3) Simple columnar (4) Simple ciliated

156. Choose the **odd** one from the options given below w.r.t. ectodermal origin.
(1) Astrocytes (2) Oligodendrocytes
(3) Microglial cells (4) Nerve cells

157. Neuroglial cells make up more than ____ percent of the volume of neural tissue in our body.
Choose the option which fills the blank **correctly**.
(1) 10 (2) 20
(3) 40 (4) 50

158. Cells which form the blood brain barrier are known as
(1) Nerve cells (2) Microglial cells
(3) Astrocytes (4) Oligodendrocytes

159. Choose the **incorrect** statement.
(1) Dendrites carry nerve impulses towards the cyton.
(2) Axon carries impulses away from the cyton.
(3) Neuroglial cells are excitable cells which transmit nerve impulses.
(4) Neuroglial cells are present in the brain and spinal cord.

160. Myelin sheath in CNS is formed by
(1) Oligodendrocytes (2) Schwann cells
(3) Microglial cells (4) Nerve cells

161. The epithelial tissue lining the inner surface of oviducts is
(1) Simple cuboidal
(2) Simple squamous
(3) Pseudostratified epithelium
(4) Simple ciliated columnar

162. Select the **correct** statement.
(1) Nissl's granules are small and regular masses of mitochondria and SER.
(2) Dendrites are long and unbranched processes arising from the axon.
(3) Nerve impulses pass between two neurons through the synapse.
(4) Complete length of axon is covered by myelin sheath in continuous manner in PNS.

163. Choose **odd** one w.r.t. properties of non-striated muscle fibres.
(1) Myofibrils show alternate light and dark bands.
(2) Fibres are taper at both ends
(3) Fibres are unbranched.
(4) They are uninucleated

164. Striated muscles are **not** located in
(1) Limbs (2) Body wall
(3) Neck (4) Wall of uterus

165. Cartilage dissolving cells are called
(1) Chondrocytes (2) Chondroclasts
(3) Chondroblasts (4) Osteoblasts

166. Blood is a fluid connective tissue which does **not** contains
(1) Thrombocytes (2) Erythrocytes
(3) Leucocytes (4) Osteocytes

167. Choose the **incorrect** one w.r.t. bones
(1) Lacunae and lamellae are absent
(2) Matrix is impregnated with salts, chiefly calcium phosphate and carbonate
(3) Growth in bone is bidirectional
(4) Osteoblasts mature to form osteocytes

168. In long bones of mammals, Haversian canals are connected to each other by
(1) Radial canals
(2) Volkmann's canals
(3) Haversian lamellae
(4) Lacuna

169. Cartilage does **not** occur in the
(1) Nasal septum (2) Larynx
(3) Skull of man (4) Tracheal rings

170. Read the following statements and choose the **correct** option.
Statement A : The matrix of cartilage is solid, pliable and resists compression.
Statement B : Osteocytes lie in lacunae in groups of two or four.
(1) Both statements are correct
(2) Both statements are incorrect
(3) Statement A is incorrect but B is correct
(4) Statement A is correct but B is incorrect

171. In adult humans, cartilage is **not** present in the
(1) Tip of nose
(2) Outer ear joints
(3) Between adjacent bones of the vertebral column
(4) Cranium

172. Nucleus is pushed to the periphery by storage of nutrients in
(1) Adipocytes (2) Fibroblasts
(3) Mast cells (4) Macrophages

173. Select the **mismatch** w.r.t. blood.
(1) WBCs – Fight against pathogen
(2) Plasma cells – Produce antigens
(3) RBCs – Transport gases from one place to another
(4) Platelets – Participate in blood clotting

174. Mast cells do **not** produce
(1) Melatonin (2) Histamine
(3) Heparin (4) Serotonin

175. Bone **not** formed through the replacement of cartilage with bony material is
(1) Clavicle (2) Humerus
(3) Femur (4) Vertebrae

176. Choose the **incorrect** statement w.r.t. bone.
(1) Non-flexible tissue
(2) Made up of nearly 70% inorganic matter and 30% organic matter
(3) If bone is put in KOH, it gets decalcified
(4) Bone marrow of long bones is the site of production of blood cells

177. Ligaments are structures which connect
(1) Skeletal muscles to bones
(2) Bone to bone
(3) Non-striated muscles to bones
(4) Cardiac muscles to bones

178. Which of the following cells produce histamine which dilates the wall of blood vessels in a allergic reactions
(1) Fibroblasts (2) Histiocytes
(3) Mast cells (4) Plasma cells

179. Read the following statements and choose the **correct** option.
Statement A : Generally new-born mammals do not shiver inspite of lower temperature outside.
Statement B : Brown fat present in them is thermogenic *i.e.* has a capacity for generating heat.
(1) Both statements are incorrect
(2) Statement B is the correct explanation of statement A
(3) Only statement A is correct
(4) Only statement B is correct

180. Tendons and ligaments are a part of
(1) Specialized connective tissue
(2) Loose connective tissue
(3) Dense irregular connective tissue
(4) Dense regular connective tissue

181. Sprain is caused due to excessive stretching of
(1) Tendons
(2) Ligaments
(3) Muscles
(4) Bones

182. Among the following, choose the epithelium which is mainly protective in function and has a limited role in secretion and absorption.
(1) Simple squamous epithelium
(2) Simple cuboidal epithelium
(3) Simple columnar epithelium
(4) Compound epithelium

183. Choose the **odd** one w.r.t. secretion of exocrine glands.
(1) Mucus (2) Enzymes
(3) Ear wax (4) Hormones

184. Choose the **incorrect** statement.
(1) Cells in an epithelial layer are held together by a large amount of intercellular material.
(2) In nearly all animal tissues, specialized junctions provide both structural and functional link between its individual cells.
(3) Hormones are secreted into the fluid bathing the gland.
(4) Endocrine glands do not have ducts.

185. Which one is **not** a multicellular gland?
(1) Mammary gland
(2) Goblet cell of the alimentary canal
(3) Salivary gland
(4) Sweat gland

SECTION-B

186. Compound epithelium is **not** located on the
(1) Moist surface of buccal cavity
(2) Inner lining of duct of salivary glands
(3) Inner lining of pancreatic duct
(4) Inner surface of trachea

187. Match column I and II. Choose the **correct** option.

	Column-I		Column-II
a.		(i)	Ciliated columnar epithelium
b.		(ii)	Simple columnar epithelium
c.		(iii)	Simple cuboidal epithelium
d.		(iv)	Simple squamous epithelium

(1) a(i), b(ii), c(iii), d(iv) (2) a(ii), b(iii), c(iv), d(i)
(3) a(iii), b(iv), c(i), d(ii) (4) a(iv), b(iii), c(ii), d(i)

188. Multicellular organism among following is
(1) *Hydra* (2) *Amoeba*
(3) *Paramecium* (4) *Euglena*

189. An example of a holocrine gland is
(1) Salivary gland (2) Sebaceous gland
(3) Sweat gland (4) Intestinal gland

190. The inner surface of the urinary bladder is lined by
(1) Transitional epithelium
(2) Stratified keratinised epithelium
(3) Columnar epithelium
(4) Squamous epithelium

191. Select the **mismatch.**
(1) Hemidesmosomes – Join epithelial cells to basal lamina
(2) Desmosomes – Perform cementing to keep the neighbouring cells together
(3) Gap junctions – Facilitate fusion of adjacent cells for movement of only big molecules.
(4) Zona adherens – Probably serve anchoring functions

192. Match column A and B. Choose the **correct** option.

Column A	Column B
a. I-band	(i) Has only myosin filaments
b. A-band	(ii) Has only actin filaments
c. H-zone	(iii) Has both actin and myosin filaments
d. Z-line	(iv) A membrane to which actin filaments are attached

(1) a(i), b(ii), c(iii), d(iv) (2) a(ii), b(iii), c(i), d(iv)

(3) a(iii), b(iv), c(ii), d(i) (4) a(iv), b(i), c(iii), d(ii)

193. Cells of squamous and cuboidal epithelium respectively are

(1) Columnar and flattened

(2) Flattened and cube-like

(3) Flattened and columnar

(4) Cube and slender with basal nucleus

194. In which type of epithelial tissue, nuclei are located closer to the base of cells?

(1) Simple columnar epithelium

(2) Simple cuboidal epithelium

(3) Simple squamous epithelium

(4) Compound cuboidal epithelium

195. Select the **mismatch.**

(1) Unipolar neurons – Have multiple dendrites

(2) Bipolar neurons – Have one axon and one dendrite

(3) Multipolar neurons – Have one long axon and remaining short processes as dendrites

(4) Pseudounipolar neurons – A single process arises first from the cell body which divides to form dendrite and axon

196. Epithelium tissue is distinguished from connective tissue, muscular and nervous tissue by:

(1) Presence of basement membrane

(2) Large amount of extracellular substance

(3) Contractibility

(4) Can carry impulses

197. Match the column-A and column-B

	Column-A		Column-B
a.	Simple ciliated epithelium	(i)	Exchange of gases between blood capillaries and tissue
b.	Simple columnar epithelium	(ii)	Help in reabsorption of substance from PCT
c.	Simple squamous epithelium	(iii)	Help in secretion, absorption from stomach and intestine.
d.	Simple cuboidal epithelium	(iv)	Found in hollow organ and help in transportation of material in specific direction

(1) a(iii), b(iv), c(ii), d(i) (2) a(iv), b(ii), c(iii), d(i)

(3) a(iv), b(iii), c(i), d(ii) (4) a(iv), b(ii), c(i), d(iii)

198. Haversian system is the feature of

(1) Cartilage

(2) Bone

(3) Loose connective tissue

(4) Tendon

199. Fusiform shaped, non-striated muscle fibres, held together by cell junctions are present in

(1) Stomach and intestinal wall

(2) Muscles of forelimb

(3) Muscles of abdominal wall

(4) Wall of heart

200. Select the **correctly** matched pair from the following options.

(1) Loose connective tissue – Adipose, lymph

(2) Fibroblast cell – Blood

(3) Dense connective tissue – Bone

(4) Chondrocytes – Cartilage

❑ ❑ ❑

03/08/2022

RM-G2
CODE-A

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MM : 720

FORTNIGHTLY TEST SERIES

(*for* NEET-2023)

Time : 3 hrs. 20 min

Test - I

Answers

1. (4)	41. (1)	81. (1)	121. (2)	161. (4)
2. (3)	42. (2)	82. (4)	122. (2)	162. (3)
3. (4)	43. (2)	83. (2)	123. (3)	163. (1)
4. (2)	44. (2)	84. (2)	124. (1)	164. (4)
5. (3)	45. (4)	85. (3)	125. (3)	165. (2)
6. (4)	46. (3)	86. (1)	126. (1)	166. (4)
7. (3)	47. (3)	87. (1)	127. (2)	167. (1)
8. (2)	48. (2)	88. (2)	128. (3)	168. (2)
9. (1)	49. (2)	89. (4)	129. (2)	169. (3)
10. (1)	50. (1)	90. (3)	130. (4)	170. (4)
11. (4)	51. (1)	91. (1)	131. (1)	171. (4)
12. (2)	52. (3)	92. (4)	132. (1)	172. (1)
13. (3)	53. (2)	93. (2)	133. (3)	173. (2)
14. (1)	54. (4)	94. (3)	134. (4)	174. (1)
15. (1)	55. (2)	95. (1)	135. (2)	175. (1)
16. (2)	56. (4)	96. (2)	136. (2)	176. (3)
17. (2)	57. (1)	97. (4)	137. (3)	177. (2)
18. (2)	58. (3)	98. (2)	138. (1)	178. (3)
19. (3)	59. (2)	99. (4)	139. (2)	179. (2)
20. (2)	60. (1)	100. (1)	140. (1)	180. (4)
21. (1)	61. (4)	101. (2)	141. (1)	181. (2)
22. (3)	62. (4)	102. (3)	142. (2)	182. (4)
23. (4)	63. (4)	103. (2)	143. (3)	183. (4)
24. (3)	64. (3)	104. (1)	144. (2)	184. (1)
25. (2)	65. (2)	105. (2)	145. (3)	185. (2)
26. (2)	66. (2)	106. (1)	146. (2)	186. (4)
27. (1)	67. (2)	107. (2)	147. (2)	187. (4)
28. (2)	68. (1)	108. (3)	148. (1)	188. (1)
29. (2)	69. (2)	109. (2)	149. (4)	189. (2)
30. (4)	70. (1)	110. (3)	150. (3)	190. (1)
31. (4)	71. (3)	111. (1)	151. (4)	191. (3)
32. (2)	72. (4)	112. (2)	152. (4)	192. (2)
33. (3)	73. (2)	113. (3)	153. (4)	193. (2)
34. (2)	74. (2)	114. (3)	154. (1)	194. (1)
35. (1)	75. (2)	115. (1)	155. (2)	195. (1)
36. (3)	76. (2)	116. (3)	156. (3)	196. (1)
37. (1)	77. (1)	117. (1)	157. (4)	197. (3)
38. (4)	78. (4)	118. (4)	158. (3)	198. (2)
39. (1)	79. (4)	119. (2)	159. (3)	199. (1)
40. (2)	80. (4)	120. (1)	160. (1)	200. (4)

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Test - I

Answers & Solutions

PHYSICS

SECTION-A

1. Answer (4)

   Gravitational force is having infinite range.

2. Answer (3)

   1 mg = 10^{-6} kg

3. Answer (4)

   TV broadcast is done using radio waves (EM waves).

4. Answer (2)

   Classical physics has concepts of nature related to macroscopic phenomenon.

5. Answer (3)

   C.V. Raman is associated with discovery of scattering of light.

6. Answer (4)

   $1^\circ = \frac{\pi}{180}$ radians and 1° = 60 minutes.

7. Answer (3)

   SI unit of force is newton which is represented as 'N'.

8. Answer (2)

   In 0.020 only last two digits are significant.

9. Answer (1)

   Velocity has SI unit ms^{-1} and dimensional formula at $[M^0LT^{-1}]$

10. Answer (1)

$$(10004)^{\frac{1}{4}} = (10000+4)^{\frac{1}{4}}$$

$$= 10\left(1+\frac{4}{10000}\right)^{\frac{1}{4}}$$

$$= 10\left(1+\frac{0.0004}{4}\right) = (1.0001)10$$

$$= 10.001$$

11. Answer (4)

    Smaller the least count, more will be the precision.

12. Answer (2)

    110.32 cm $\approx$ 110 cm upto three significant figure.
    In scientific rotation

    110 cm = 1.10×10^2 cm

13. Answer (3)

$$P = \frac{W}{t} = \frac{[ML^2T^{-2}]}{[T]} = [ML^2T^{-3}]$$

14. Answer (1)

    2A – B = (20.0 ± 0.2) – (5.0 ± 0.1)

    = (15.0 ± 0.3)

15. Answer (1)

    SI unit of electric current is ampere (A)

16. Answer (2)

— 100 3 — 100 $\frac{1}{2}$ — 100

$\frac{1}{2}$ — 100

– –

17. Answer (2)

If number is less than 1, then zeros between two non-zero digits and trailing zeros are significant.

18. Answer (2)

1.4 + 4.24 + 7.432 = 13.072 = 13.1 g

Hence, result has 3 significant figures.

19. Answer (3)

$T = (T_2 - T_1) \pm (T_1 + T_2)$

$= (44 - 31) \pm (1 + 2)°C$

$= (13 \pm 3)°C$

20. Answer (2)

Zero error is classified under systematic error.

21. Answer (1)

Distance and focal length, both have dimensional formula $[M^0LT^0]$.

22. Answer (3)

Magnification is a dimensionless constant.

23. Answer (4)

'kx' and ' t' both should be dimensionless

dimensionless [ ] = $[T^{-1}]$

dimensionless [k] = $[L^{-1}]$

–

k

24. Answer (3)

1 J = 10^7 ergs

100 J = 10^9 ergs

25. Answer (2)

L = (velocity) (time)

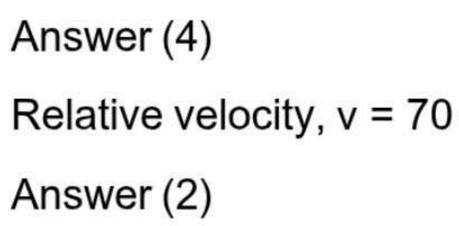
L = uT

$[L] = [F^0uT]$

26. Answer (2)

Slope $\frac{dx}{dt} = v$ (Velocity)

27. Answer (1)



$|\vec{S}|$ = – =

28. Answer (2)

× — =

29. Answer (2)

$y = x^3$

—

30. Answer (4)

Slope of A i.e. velocity is constant.

Its acceleration is zero.

31. Answer (4)

Relative velocity, v = 70 – 70 = 0

32. Answer (2)

In (2), slope of P_1 > slope of P_2

$v_1 > v_2$

33. Answer (3)

$$\langle \ \rangle = \frac{\int}{\int} = \frac{\int \alpha}{\quad}$$

34. Answer (2)

$$- \frac{^1\frac{-}{5} \quad ^2\frac{4}{5}}{\quad}$$

$$= \frac{^1 + 4 \ ^2}{5}$$

35. Answer (1)

Instantaneous speed is equal to magnitude of instantaneous velocity.

SECTION-B

36. Answer (3)

Distance travelled = 20 + 0 + 20 = 40 m

37. Answer (1)

S S –gt t –gt

100 = 50t

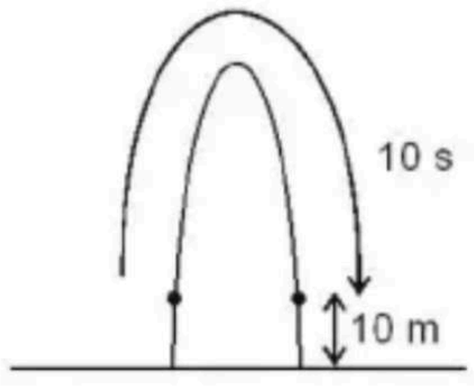
t = 2 s

38. Answer (4)

— $\frac{2}{5}$

i.e. a depends on time.

39. Answer (1)



10 $\frac{1}{2}$ 10 5^2 135 m

$\sqrt{\frac{2}{}}$ $\sqrt{\frac{2\ 135}{10}}$ $\sqrt{\ }$ $\sqrt{\ }$

T t $\sqrt{\ }$

40. Answer (2)

–

5 $\frac{2}{2}$(10 1) 9 m

41. Answer (1)

$x = 2 + 12t – t^2$

$v = 12 – 2t$

v = 0 t = 6 s

$a = –2\ m/s^2$

Retardation = $2\ m/s^2$

42. Answer (2)

x = area under the v-t curve.

= – × × =

43. Answer (2)

$6^{\ 2}$ 6 4

12 6

a t

44. Answer (2)

Instantaneous velocity, — = slope of x – t graph

— is negative at 'c'

45. Answer (4)

At a given time, only one value of velocity is possible.

46. Answer (3)

P F e ^{t}

Exponential power are dimensionless

$[t] = [M^0L^0T^0]$

$= [M^0L^0T^{-1}]$

$[P] = [F]$

$[ML^{-1}T^{-2}] = [MLT^{-2}]$

$\frac{[MLT^{\ 2}][T^{\ 1}]}{[ML^{\ 1}T^{\ 2}]\ [L^2T^{\ 1}]}$ $[L^2T^{\ 1}]$

— ——

47. Answer (3)

— ——

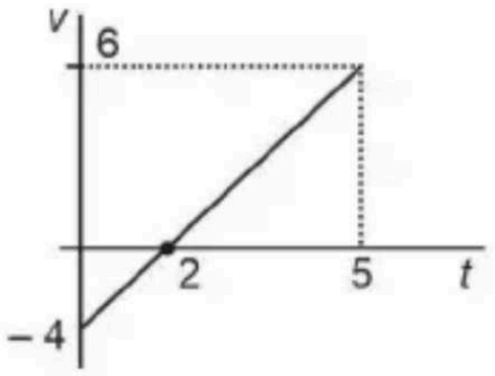
— $\frac{m}{m}$ $\frac{V}{V}$

— — ——

$= \left(\text{—} + \text{—}\right) = \frac{+}{\text{——}}$

$= \frac{240}{1000} = 0.24$

48. Answer (2)



Distance $= \frac{1}{2}(4)(2) + \frac{1}{2}(3)(6) = 4 + 9 = 13\ m$

49. Answer (2)

The particle acquires same speed after returing to same level

Their velocity of strike will be same

But, $m_A > m_B$

$(K.E)_A > (K.E.)_B$

50. Answer (1)

Let length of each train be 'ℓ'

Then, ℓ — –

$2\ell = -5(20) + (20)^2$

$2\ell = -100 + 400$

$\boxed{\ell = 150\ m}$

CHEMISTRY

SECTION-A

51. Answer (1)

	A	+ 2B	C
Initial	5 mole	12 mol	0
Final	0	2 mole	5 mole

A is limiting reagent

Maximum moles of C = 5 mole

52. Answer (3)

Number of O-atoms —

53. Answer (2)

Mole of H_2SO_4 = ——

Mass of solution = 100 g

Volume of solution —

Molarity = —— × ——

$= \frac{24.5}{98} \times \frac{1.2}{100} \times 1000$

Molarity = 3 M

54. Answer (4)

Molality ——

55. Answer (2)

0.1308 g of zinc is present in 100 g of enzyme

1 g of zinc is present in —— of enzyme

65.4 g of zinc is present in $\frac{100 \quad 65.4}{0.1308}$ g of enzyme

= 50000 u

56. Answer (4)

Mass % of oxygen = —— × =

57. Answer (1)

N = M × n-factor

58. Answer (3)

$2NO + O_2 \quad 2NO_2$ at STP.

2 L 1 L 2 L

59. Answer (2)

No. of P-atoms = 0.5 mol × 4 × N_A = 2 N_A

60. Answer (1)

Molecules of H_2 – × N_A = 2 N_A

Molecules of $N_2 = \frac{28}{28} \times N_A = N_A$

Molecules of O_2 = 0.5 N_A

Molecules of — —

61. Answer (4)

Molarity of $CaCl_2$ solution = —— =

Molarity of chloride ions = 0.5 × 2 = 1 M

62. Answer (4)

Zeros preceding to first non-zero digit are not significant.

63. Answer (4)

Atoms are neither created nor destroyed in a chemical reaction.

64. Answer (3)

When two different elements combine to form different compounds then law of multiple

proportions is applicable. CS_2 and CO_2 will not follow law of multiple proportions.

65. Answer (2)

Mass of one molecule ——————

66. Answer (2)

Mole of carbon —

Mole of hydrogen —

Mole ratio of C : H = 6.66 : 20 = 1 : 3

Empirical formula = CH_3

67. Answer (2)

Mole of methane —— mol

Mole of atoms of methane = 0.1 × 5 6.02×10^{23}

$= 3.01 \times 10^{23}$

68. Answer (1)

SI unit of mass is kg

69. Answer (2)

Gay Lussac's law of gaseous volume is applicable when reactants and products are in gas phase.

70. Answer (1)

The concentration term which does not contain volume term is independent of temperature.

71. Answer (3)

————— ———

——

72. Answer (4)

Average atomic mass

————————— amu

73. Answer (2)

Volume = mass × density = 18 × 1 = 18 ml

74. Answer (2)

$M_1V_1 = M_2V_2$

$0.2 \times V_1 = 0.1 \times 1000$ $V_1 = 500$ ml

75. Answer (2)

Number of electrons $\frac{1.2}{16}$ 10 N_A $0.75\ N_A$

76. Answer (2)

$CH_4 + 2O_2$ $CO_2 + 2H_2O$

1 mole CH_4 requires 2 mole of O_2 for complete reaction.

Volume of O_2 required at STP = 22.4 × 2 = 44.8 L

77. Answer (1)

Mole of H_2O in 1 L H_2O —— mol

Molarity of H_2O = 55.55 M

78. Answer (4)

$$= \frac{\text{—}}{\text{—} + \text{—}} = \text{—}$$

79. Answer (4)

Number of Be atoms —

80. Answer (4)

A given compound always contains exactly the same proportion of elements by weight. This is called law of definite proportions.

81. Answer (1)

In case of ionic compounds formula mass is preferred to molecular mass.

82. Answer (4)

Molecular formula shows the exact number of different types of atoms present in a molecule of a compound.

83. Answer (2)

Moles of $NO_x = \frac{30}{14+16x} = \frac{22.4}{22.4}$

x = 1

84. Answer (2)

85. Answer (3)

$\frac{n_{O_2}}{n_{H_2}}$ $\frac{\frac{W}{32}}{\frac{W}{2}}$ $\frac{1}{16}$

SECTION-B

86. Answer (1)

Generally isotopes of an element have different relative abundance.

87. Answer (1)

1 amu = 1.66×10^{-27} kg

88. Answer (2)

Molecular formula = $CH_3COOH = C_2H_4O_2$

Empirical formula = CH_2O

89. Answer (4)

$2HCl + CaCO_3 \quad CaCl_2 + CO_2 + H_2O$

Mole of $CaCO_3$ $\frac{50}{100}$ 0.5

Mole of HCl required = 0.5 × 2 = 1

Mass of HCl required = 36.5 g

25 g HCl is present in 100 g of HCl

36.5 g HCl present in $\frac{100}{25}$ 36.5

= 36.5 × 4 g

90. Answer (3)

Moles of oxygen atoms = 0.1 × 14 = 1.4 mol

91. Answer (1)

Compound	Mass% of carbon
C_6H_6	—
C_2H_6	—
C_2H_4	—
C_3H_4	—

92. Answer (4)

Mass of HNO_3 is the solution = $200 \times 3 \times 63 \times 10^{-3}$

= 37.8 g

70 g of HNO_3 is present in 100 g of conc. NHO_3

1 g of HNO_3 is present in —

37.8 g of NHO_3 is present in $\frac{100}{70}$ 37.8

= 54 g

93. Answer (2)

Molarity (M) —

94. Answer (3)

$\xrightarrow{\Delta}$ +

2.2 g CO_2 is formed by

$= \frac{84}{44} \times 2.2 = 4.2$ g $MgCO_3$

Percentage purity —

95. Answer (1)

—

$n_{NaCl} = \frac{5.85}{58.5} = 0.1$ mol

Mass of AgCl formed = 0.1 × 143.5

= 14.35 g

96. Answer (2)

–

–

° = – ×–

°C = 65.56

97. Answer (4)

$CaCO_3 + 2HCl \quad CaCl_2 + CO_2 + H_2O$

Mole of HCl = $200 \times 4 \times 10^{-3} = 8 \times 10^{-1} = 0.8$

Mole of $CaCO_3$ reacted —

Mass of $CaCO_3 = 0.4 \times 100 = 40$ g

98. Answer (2)

1 L = 1000 cm^3

= $(1000 \times 10^{-2} \times 10^{-2} \times 10^{-2})$ m^3

1 L = 10^{-3} m^3

20 L = 20×10^{-3} m^3

= 2×10^{-2} m^3

99. Answer (4)

Multiple	Prefix
10^{-9}	Nano
10^{-15}	Femto
10^{-6}	Micro
10^{-12}	Pico

100. Answer (1)

3.004 has four significant figures

0.0041 has two significant figures

BOTANY

SECTION-A

101. Answer (2)

Microbodies are present in both plant and animal cells.

102. Answer (3)

Satellite chromosomes have secondary constriction while sub-metacentric chromosome has centromere slightly away from the middle of the chromosome.

103. Answer (2)

Chromatin contains DNA, some basic proteins called histones, some non-histone proteins and RNA.

104. Answer (1)

Nucleolus is the site of rRNA synthesis.

105. Answer (2)

Two centrioles are present in a centrosome.

106. Answer (1)

The basal body of cilia and flagella have centriole like arrangement of microtubules (9 + 0 arrangement)

107. Answer (2)

Cilia works like oars.

108. Answer (3)

Prokaryotic flagella is structurally different from eukaryotic flagella.

109. Answer (2)

Anton von Leeuwenhoek first saw a living cell.

110. Answer (3)

Ribosomes are non-membrane bound organelles.

111. Answer (1)

Mycoplasma are smallest cell of 0.3 m in length.

The longest isolated single cell is the egg of ostrich.

Nerve cells are the longest cells.

112. Answer (2)

A network of filamentous proteinaceous structures present in the cytoplasm is collectively referred to as cytoskeleton.

113. Answer (3)

The cell theory states that

All living organism are composed of cells and products of cells.

All cells arise from pre-existing cells.

114. Answer (3)

The main constituent of fungal cell wall is chitin.

115. Answer (1)

PPLO is a prokaryotic cell.

116. Answer (3)

Those bacteria that take up the gram stain are gram positive and others that do not are called gram negative bacteria.

117. Answer (1)

Glycocalyx is a loose sheath called slime layer.

Glycocalyx may be thick and tough called capsule

118. Answer (4)

Mesosomes do not perform the function of photosynthesis.

119. Answer (2)

Chromatophores are found in cyanobacteria.

120. Answer (1)

The fimbriae are small bristle sprouting out of the cell and they are known to help attach the bacteria to rocks in streams and also to the host tissue.

121. Answer (2)

Several ribosomes may attach to a single m-RNA and form a chain called polysome.

122. Answer (2)

Inclusion bodies are non-membrane bounded structures that lie free in the cytoplasm.

123. Answer (3)

Cytoskeleton is present in eukaryotes and genetic material is organised into chromosomes in them.

124. Answer (1)

The lipid component of the membrane mainly consist of phosphoglycerides.

125. Answer (3)

Depending on the ease of extraction, membrane proteins can be classified as integral or peripheral.

126. Answer (1)

According to fluid mosaic model, the quasi fluid nature of the lipid enables lateral movement of proteins within overall bilayer. This ability to move within the membrane is measured as its fluidity.

127. Answer (2)

When molecule move across membrane along the concentration gradient it is called passive transport.

Active transport is movement of ion against concentration gradient.

Movement of water by diffusion is called osmosis.

128. Answer (3)

Regulation of transport of materials is the function of cell membrane.

129. Answer (2)

Middle lamella is mainly made up of calcium pectate.

130. Answer (4)

ER and Golgi body are the part of endomembrane system and their functions are coordinated.

131. Answer (1)

RER is involved in protein synthesis and secretion.

132. Answer (1)

Steroidal hormones are synthesized in SER.

133. Answer (3)

Centromere appear at terminal position and chromosome appear to have only one arm in telocentric chromosome.

134. Answer (4)

Microfilament help in pseudopodia formation.

135. Answer (2)

Golgi apparatus is related to the processing, packaging and transport of material.

SECTION-B

136. Answer (2)

The hydrolytic enzymes found in the isolated lysosome are optimally active at the acidic pH.

137. Answer (3)

The sap-vacuole is bound by a single membrane called tonoplast.

138. Answer (1)

Mitochondria are the site of aerobic respiration.

139. Answer (2)

Chloroplast - green coloured

Chromoplast - coloured other than green

Leucoplasts - colourless.

140. Answer (1)

Carotenoids are hydrophobic and are fat soluble pigments.

141. Answer (1)

Amyloplasts store carbohydrates, elaioplasts store oils and fats and aleuroplasts store proteins. Amyloplasts, elaioplasts, aleuroplasts are types of leucoplasts.

142. Answer (2)

The space limited by inner membrane of the chloroplast is called the stroma.

143. Answer (3)

In cilia and flagella 9 peripheral doublets and 2 microtubules are found in centre (9 + 2 arrangement)

144. Answer (2)

In Amoeba, the contractile vacuole is important for excretion.

145. Answer (3)

Plasmid is a small single circular chromosomal DNA present in addition to the genomic DNA

146. Answer (2)

Mesosomes are specialised differentiated form of cell membrane in prokaryotes.

147. Answer (2)

In prokaryotic cell, flagella helps in motility.

148. Answer (1)

Ribosome is found in cytoplasm but also within the cell organelle such as mitochondria and chloroplast.

149. Answer (4)

Animal cells have centrioles which are absent in almost all plant cells.

150. Answer (3)

A few chromosome have non-staining secondary constriction at a constant location. This gives the appearance of small fragment called satellite.

ZOOLOGY

SECTION-A

151. Answer (4)

When two or more organs perform a common function by their physical and/or chemical interaction, they together form an organ system e.g., digestive system, respiratory system.

152. Answer (4)

Tissues are broadly classified into four types

(i) Epithelial (ii) Connective (iii) Muscular and (iv) Neural

153. Answer (4)

Tight junctions provide a barrier which prevent leakage of substances or fluids across epithelial cells.

Adhering junctions aid to cement adjacent cells.

Gap junctions provide cytoplasmic connections to facilitate communication between adjacent cells.

Synaptic junctions help in transmission of information through chemicals.

154. Answer (1)

Bronchioles and fallopian tubes are lined with ciliated epithelium to move particles or mucus in a specific direction.

155. Answer (2)

Squamous epithelial cells closely fit like the tiles in a floor. Hence, it is also known as pavement epithelium.

156. Answer (3)

Microglial cells are mesodermal in origin. They are small in size with few feathery processes and help in phagocytosis.

157. Answer (4)

About more than 50% i.e. one half the volume of neural tissues of brain cells are neuroglial cells.

158. Answer (3)

Astrocytes help in repair of neural tissue and form blood brain barrier.

159. Answer (3)

Neurons are excitable cells which transmit nerve impulses.

160. Answer (1)

Oligodendrocytes are with a few protoplasmic processes and form myelin sheath in CNS.

161. Answer (4)

Ciliated columnar epithelium consists of columnar cells which have cilia on their free surfaces. This epithelium occurs in the inner surface of hollow organs like fallopian tube (oviducts) and most of the respiratory tract.

162. Answer (3)

Nissl's granules are large and irregular masses of ribosomes and RER.

Dendrites are short and branched processes arising from the cyton.

Axon is surrounded by a sheath called neurilemma.

163. Answer (1)

Myofibrils of non-striated muscles lack light and dark bands.

164. Answer (4)

Non-striated muscles occur in the posterior part of oesophagus, urinogenital tract, iris of eye etc.

165. Answer (2)

Osteoclasts are bone dissolving cells.

166. Answer (4)

Blood is a fluid connective tissue containing plasma, erythrocytes, leucocytes and platelets.

Osteocytes are bone cells.

167. Answer (1)

In bone, lacunae and lamellae are present.

168. Answer (2)

Compact bone is composed of many parallel, longitudinal column-like structures called Haversian system, cemented to each other. Haversian canals are connected to each other by Volkmann's canals.

169. Answer (3)

In hyaline cartilage, the matrix is apparently fibre-less and glass-like (hyaline) but translucent. It occurs in the larynx, nasal septum, tracheal rings and costal cartilage.

Yellow fibrocartilage/elastic cartilage forms the Eustachian tube, epiglottis and pinna of ear.

170. Answer (4)

Chondrocytes lie singly or in groups of two or four.

171. Answer (4)

Cartilage is not present in cranium.

Bone forms the endoskeleton of adult vertebrates.

172. Answer (1)

Each adipocyte contains a large droplet of fat that almost fills it.

173. Answer (2)

Plasma cells produce antibodies.

174. Answer (1)

Melatonin is a hormone.

175. Answer (1)

Skull bones and clavicle are membrane/ investing/dermal bones.

Humerus, femur, vertebrae, ribs, girdle bones except clavicle are the examples of cartilaginous bone/endochondrial/replacing bones.

176. Answer (3)

If bone is put in HCl it gets decalcified and becomes soft and flexible but nothing happens to it, if it is put in KOH.

177. Answer (2)

Ligaments connect bone to bone while tendons join muscles to bones.

178. Answer (3)

Mast cells produce histamine which dilate the wall of blood vessels in inflammatory and allergic reaction.

179. Answer (2)

Brown fat is particularly found in new-born babies and hibernating mammals.

180. Answer (4)

In the dense regular connective tissue, the collagen fibres are present in rows between many parallel bundles of fibres. e.g. tendons and ligaments.

181. Answer (2)

Sprain is caused due to excessive pulling (stretching) of ligaments.

182. Answer (4)

Compound epithelium is made of more than one layer of cells.

183. Answer (4)

Exocrine glands secrete mucus, saliva, earwax, oil, milk, digestive enzymes and other cell products.

184. Answer (1)

All cells in epithelium are held together with little intercellular material and cell junctions.

185. Answer (2)

Goblet cells are unicellular glands which secrete mucus.

SECTION-B

186. Answer (4)

Ciliated epithelium is present in the inner surface of trachea.

187. Answer (4)

Squamous epithelium is made of a single thin layer of flattened cells with irregular boundaries.

188. Answer (1)

Hydra is an example of multicellular animal.

189. Answer (2)

In holocrine glands, the whole cell is destroyed while releasing the product.

190. Answer (1)

Transitional epithelium allows considerable expansion of the urinary bladder to accommodate urine, because stretching considerably flattens and broadens the cells of superficial and middle layers.

191. Answer (3)

Interdigitations are interfitting finger-like processes of the cell membranes of adjacent cells. They increase the surface area.

192. Answer (2)

Each segment of the myofibril from one Z line to the next, functions as a contractile unit and is called a sarcomere.

193. Answer (2)

The squamous epithelium is made of a single thin layer of flattened cells with irregular boundaries.

194. Answer (1)

The columnar epithelium is composed of a single layer of tall and slender cells.

195 Answer (1)

Unipolar neurons have only axons but no dendrites and are found in early embryos.

196. Answer (1)

Epithelium tissues are compactly packed with little intercellular matrix. Epithelium lie on the non-cellular layer called basement membrane.

197. Answer (3)

Squamous epithelium – Found in the wall of blood vessels, helps in filtration and diffusion of gases from blood to tissue.

198. Answer (2)

Haversian system is the feature of bone.

199. Answer (1)

Smooth muscle fibres taper at both end (fusiform) and non-striated. They are present in wall of hollow visceral organs like stomach, intestine, urinary bladder, etc.

200. Answer (4)

Loose connective tissue – Areolar and adipose

Fibroblast – Not found in blood.

Dense connective – Tendon and ligament are the example of dense regular connective tissues.

❑ ❑ ❑